This is CS50

This is CS50

- We'll take a break partway through class, with Cheez-Its for snacks! 🧀
- Looking for a study buddy? Submit cs50.ly/studybuddy.
- To attend CS50 Lunch this Fri 9/19 at 1:15pm, RSVP at cs50.ly/lunch!
  - Aok to attend again this week even if you attended last week's!
- Do say hi or ask questions during break or after class!
  - What do today's binary bulbs spell?
- Happy birthday to Brenda! 🥳

This is CS50

reading levels

One fish. Two fish. Red fish. Blue fish.

Before Grade 1

Congratulations! Today is your day. You're off to Great Places! You're off and away!

Grade 3

It was a bright cold day in April, and the clocks were striking thirteen. Winston Smith, his chin nuzzled into his breast in an effort to escape the vile wind, slipped quickly through the glass doors of Victory Mansions, though not quickly enough to prevent a swirl of gritty dust from entering along with him.

Grade 10

reading levels

cryptography

U I J T J T D T 5 0

T H I S I S C S 5 0

CS50

debugging

GRACE HOPPER

IBM

Photo # NH 96566-KN (Color) First Computer "Bug", 1947

9/9

0800 Arctan started { 1.2700 9.037 847 025

1000 " stopped - arctan ✓ 9.037 846 995 correct

13 uc (033) MP - MC ~~1.5821649000~~ ~~2.130476415 (-03)~~ 4.615925059(-2)

(033) PRO 2 2.130476415

correct 2.130676415

Relays 6-2 in 033 failed special speed test

in relay " 11.00 test.

Relays changed

Relay 2145 Relay 3370

1100 Started Cosine Tape (Sine check)

1525 Started Mult + Adder Test.

1545 Relay #70 Panel F (moth) in relay.

First actual case of bug being found.

~~1645~~ 1630 Arctangent started.

1700 closed down.

In relay " 11.00 test.

Relays changed

1100 Started Cosine Tape (Sine check)

1525 Started Mult + Adder Test.

1545

Relay #70 Panel F
(moth) in relay.

First actual case of bug being found.

~~1635~~ 1630 Antangent started.

1700 closed down.

printf

printf

debug50

`printf`

`debug50`

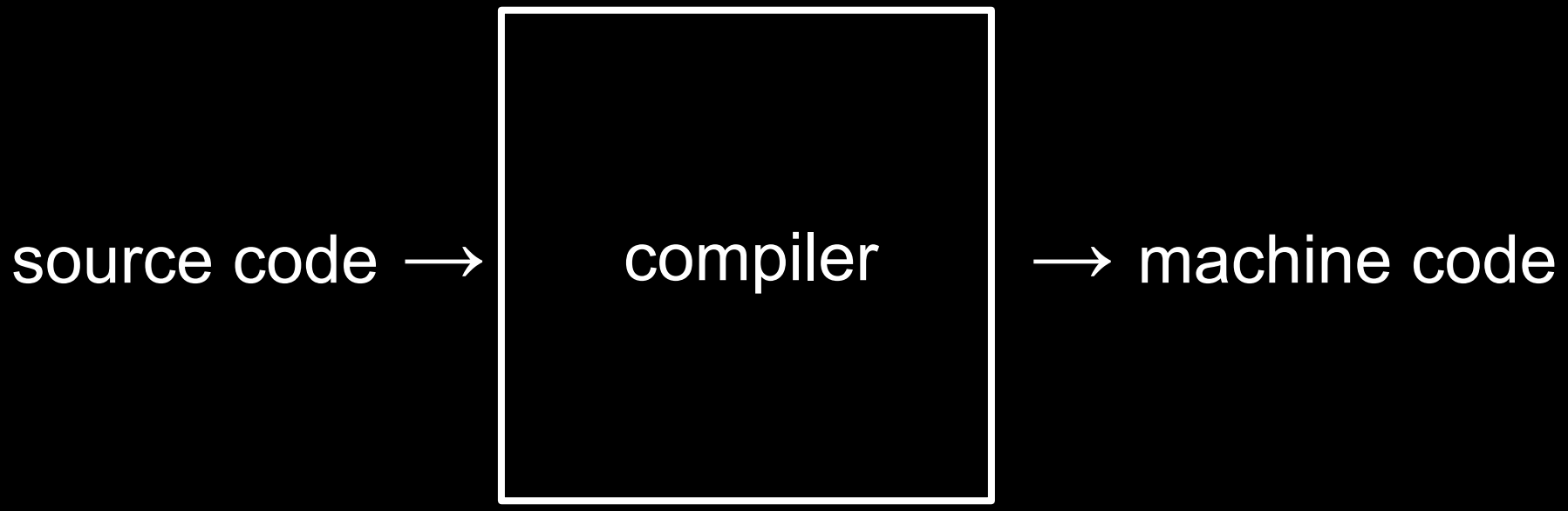
rubber duck

`printf`

`debug50`

rubber duck

CS50 Duck


source code
compiler
machine code

```c
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

```
01111111 01000101 01001100 01000110 00000010 00000001 00000001 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000010 00000000 00111110 00000000 00000001 00000000 00000000 00000000
10110000 00000101 01000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
11010000 00010011 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 01000000 00000000 00111000 00000000
00001001 00000000 01000000 00000000 00100100 00000000 00100001 00000000
00000110 00000000 00000000 00000000 00000101 00000000 00000000 00000000
01000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 00000000
01000000 00000000 01000000 00000000 00000000 00000000 00000000 00000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 00000000
11111000 00000001 00000000 00000000 00000000 00000000 00000000 00000000
00001000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000011 00000000 00000000 00000000 00000100 00000000 00000000 00000000
00111000 00000010 00000000 00000000 00000000 00000000 00000000 00000000
...
```

```
make hello

./hello
```

```
clang hello.c

./a.out
```

command-line arguments

```
clang -o hello hello.c

./hello
```

```c
#include <stdio.h>


int main(void)
{
    printf("hello, world\n");
}
```

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
clang -o hello hello.c -lcs50

./hello
```

```
make hello

./hello
```

compiling

preprocessing

compiling

assembling

linking

preprocessing

compiling

assembling

linking

```c
#include <stdio.h>

void meow(void);

int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        meow();
    }
}

void meow(void)
{
    printf("meow\n");
}
```

```c
#include <stdio.h>

void meow(void);

int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        meow();
    }
}

void meow(void)
{
    printf("meow\n");
}
```

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
string get_string(string prompt);
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
string get_string(string prompt);
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
string get_string(string prompt);
int printf(string format, ...);

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

preprocessing

compiling

assembling

linking

```
string get_string(string prompt);
int printf(string format, ...);

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq   %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```

preprocessing

compiling

assembling

linking

```
...
main:                                   # @main
    .cfi_startproc
# BB#0:
    pushq    %rbp
.Ltmp0:
    .cfi_def_cfa_offset 16
.Ltmp1:
    .cfi_offset %rbp, -16
    movq    %rsp, %rbp
.Ltmp2:
    .cfi_def_cfa_register %rbp
    subq    $16, %rsp
    xorl    %eax, %eax
    movl    %eax, %edi
    movabsq    $.L.str, %rsi
    movb    $0, %al
    callq    get_string
    movabsq    $.L.str.1, %rdi
    movq    %rax, -8(%rbp)
    movq    -8(%rbp), %rsi
    movb    $0, %al
    callq    printf
    ...
```

```
01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000001111100000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000000100000000000000
00001010000000000000000100000000
01010101010010001000100111100101
01001000100000111110110000010000
00110001110000001000100111000111
01001000101111100000000000000000
00000000000000000000000000000000
00000000000000001011000000000000
11101000000000000000000000000000
00000000010010001011111100000000
00000000000000000000000000000000
00000000000000000000000001001000
...
```

preprocessing

compiling

assembling

linking

```
#include <cs50.h>
#include <stdio.h>

int main(void)
{
    string name = get_string("What's your name? ");
    printf("hello, %s\n", name);
}
```

hello.c

hello.c

cs50.c

hello.c

cs50.c

stdio.c

```
01111111010001010100110001000110
00000010000000010000000100000000
00000000000000000000000000000000
00000000000000000000000000000000
00000001000000000011111000000000
00000001000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
10100000000000100000000000000000
00000000000000000000000000000000
00000000000000000000000000000000
01000000000000000000000000000000
00000000000000000100000000000000
00001010000000000000000100000000
01010101010010001000100111100101
01001000100000111110110000010000
00110001110000001000100111000111
01001000101111100000000000000000
00000000000000000000000000000000
00000000000000001011000000000000
11101000000000000000000000000000
00000000010010001011111100000000
00000000000000000000000000000000
00000000000000000000000001001000
...
```

cs50.c

stdio.c

```
01111111010001010100110001000110   01111111010001010100110001000110
00000010000000010000000100000000   00000010000000010000000100000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000001000000000011111000000000   00000011000000000011111000000000
00000001000000000000000000000000   00000001000000000000000000000000
00000000000000000000000000000000   11000000000011110000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   01000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
10100000000001000000000000000000   00101000001100100000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
01000000000000000000000000000000   01000000000000000011100000000000
00000000000000001000000000000000   00000111000000000100000000000000
00001010000000000000000100000000   00011100000000000001100100000000
01010101010010001000100111100101   00000001000000000000000000000000
01001000100000111110110000010000   00000101000000000000000000000000
00110001110000001000100111000111   00000000000000000000000000000000
01001000101111100000000000000000   00000000000000000000000000000000
00000000000000000000000000000000   00000000000000000000000000000000
00000000000000001011000000000000   00000000000000000000000000000000
11101000000000000000000000000000   00000000000000000000000000000000
00000000100100010111111100000000   00000000000000000000000000000000
00000000000000000000000000000000   01011100001001010000000000000000
00000000000000000000000001001000   00000000000000000000000000000000
...                                ...
```

stdio.c

```
011111110100010101001100010001100
000000100000000010000000100000000
000000000000000000000000000000000
000000000000000000000000000000000
000000010000000000111110000000000
000000001000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
101000000000001000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
010000000000000000000000000000000
000000000000000000010000000000000
000010100000000000000000100000000
010101010100100010001001111001 01
010010001000001111101100000100 00
001100011100000010001001110001 11
010010001011111000000000000000000
000000000000000000000000000000000
000000000000000010110000000000000
111010000000000000000000000000000
000000000100100010111111000000000
000000000000000000000000000000000
000000000000000000000000001001000
...
```

```
011111110100010101001100010001100
000000100000000010000000100000000
000000000000000000000000000000000
000000000000000000000000000000000
000000011000000000011111000000000
000000001000000000000000000000000
110000000000111100000000000000000
000000000000000000000000000000000
010000000000000000000000000000000
000000000000000000000000000000000
001010000011001000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
010000000000000000011100000000000
000001110000000000100000000000000
000111000000000000011001000000000
000000010000000000000000000000000
000001010000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
000000000000000000000000000000000
010111000010010100000000000000000
000000000000000000000000000000000
...
```

```
001011110110110001101001011000 10
011000110010111001110011011011 11
001011100011011000100000001011 11
011101010111001101110010001011 11
011011000110100101100010001011 11
011110000011100000110110010111 11
001101100011010000101101011011 00
011010010110111001110101011110 00
001011010110011101101110011101 01
001011110110110001101001011000 10
011000110101111101101110011011 11
011011100111001101101000011000 01
011100100110010101100100001011 10
011000010010000000100000010000 01
010100110101111101001110010001 01
010001010100010001000101010001 00
001000000010100000100000001011 11
011011000110100101100010001011 11
011110000011100000110110010111 11
001101100011010000101101011011 00
011010010110111001110101011110 00
001011010110011101101110011101 01
001011110110110001100100001011 01
011011000110100101101110011101 01
011110000010110101111000001110 00
001101100010110100110110001101 00
...
```

0111111101000101010011000100011000000100000000100000001000000000000000000000000000000000000000000000
0000000000000000000000000000000100000000001111100000000000000100000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000101000000000001000000000000000000000000000000000000000000000000000000000000000000000000
0000000001000000000000000000000000000000000000000000000100000000000000000101000000000000000001000000
0001010101010010001000100111100101010010001000001111101100000100000011000111000000100010011100011101 00
1000101111100000000000000000000000000000000000000000000000000000000000000000000000000101100000000000111010000 0
0000000000000000000000000000001001000101111110000000000000000000000000000000000000000000000000000000
0000000001001000...0111111101000101010011000100011000000100000000100000001000000000000000000000000000
0000000000000000000000000000000000000000000000011000000000011111000000000000000100000000000000000
0000000110000000001111000000000000000000000000000000000000000000000000000000000100000000000000000000000
0000000000000000000000000000000000101000001100100000000000000000000000000000000000000000000000000000
0000000000000000000000000001000000000000000011100000000000000001110000000001000000000000000011100000
0000000011001000000000000000100000000000000000000000000101000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000101110000100101000000000000
0000000000000000000000000000000000...0010111110110110001101001011000100110001100101110011100110110111 1
0010111100011011000100000001011110111010101110011011100100010111101101100011010010110001000101111011110
0000111000001101100101111100110110001101000010110101101100011010010110111001110101011110000010110101 10
0111011011100111010100101111011011000110100101100010011000110101111101101110011011110110111001110011 01
1010000110000101110010011001010110010000101110011000010010000000100000010000010101001101011111010011 10
0100010101000101010001000100010101000100001000000010100000100000001011110110110001101001011000100010 11
1101111000001110000011011001011111001101100011010000101101011011000110100101101110011101010111100000 10
1101011001110110111001110101001011110110110001100100001011010110110001101001011011100111010101111000 00
1011010111100000111000001101100010110100110110001101 00...

preprocessing

compiling

assembling

linking

compiling

decompiling

reverse engineering

0111111101000101010011000100011000000100000000100000001000000000000000000000000000000000000000000000
0000000000000000000000000000000100000000001111100000000000000100000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000101000000000001000000000000000000000000000000000000000000000000000000000000000000000000
0000000001000000000000000000000000000000000000000000000100000000000000000101000000000000000001000000
0001010101010010001000100111100101010010001000001111101100000100000011000111000000100010011100011101 00
1000101111100000000000000000000000000000000000000000000000000000000000000000000000000101100000000000111010000 0
0000000000000000000000000000001001000101111110000000000000000000000000000000000000000000000000000000
0000000001001000...0111111101000101010011000100011000000100000000100000001000000000000000000000000000
0000000000000000000000000000000000000000000000011000000000011111000000000000000100000000000000000
0000000110000000001111000000000000000000000000000000000000000000000000000000000100000000000000000000000
0000000000000000000000000000000000101000001100100000000000000000000000000000000000000000000000000000
0000000000000000000000000001000000000000000011100000000000000001110000000001000000000000000011100000
0000000011001000000000000000100000000000000000000000000101000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000101110000100101000000000000
0000000000000000000000000000000000...0010111110110110001101001011000100110001100101110011100110110111 1
0010111100011011000100000001011110111010101110011011100100010111101101100011010010110001000101111011110
0000111000001101100101111100110110001101000010110101101100011010010110111001110101011110000010110101 10
0111011011100111010100101111011011000110100101100010011000110101111101101110011011110110111001110011 01
1010000110000101110010011001010110010000101110011000010010000000100000010000010101001101011111010011 10
0100010101000101010001000100010101000100001000000010100000100000001011110110110001101001011000100010 11
1101111000001110000011011001011111001101100011010000101101011011000110100101101110011101010111100000 10
1101011001110110111001110101001011110110110001100100001011010110110001101001011011100111010101111000 00
1011010111100000111000001101100010110100110110001101 00...

types

bool

int

long

float

double

char

string

...

`bool`	1 byte
`int`	4 bytes
`long`	8 bytes
`float`	4 bytes
`double`	8 bytes
`char`	1 byte
`string`	? bytes

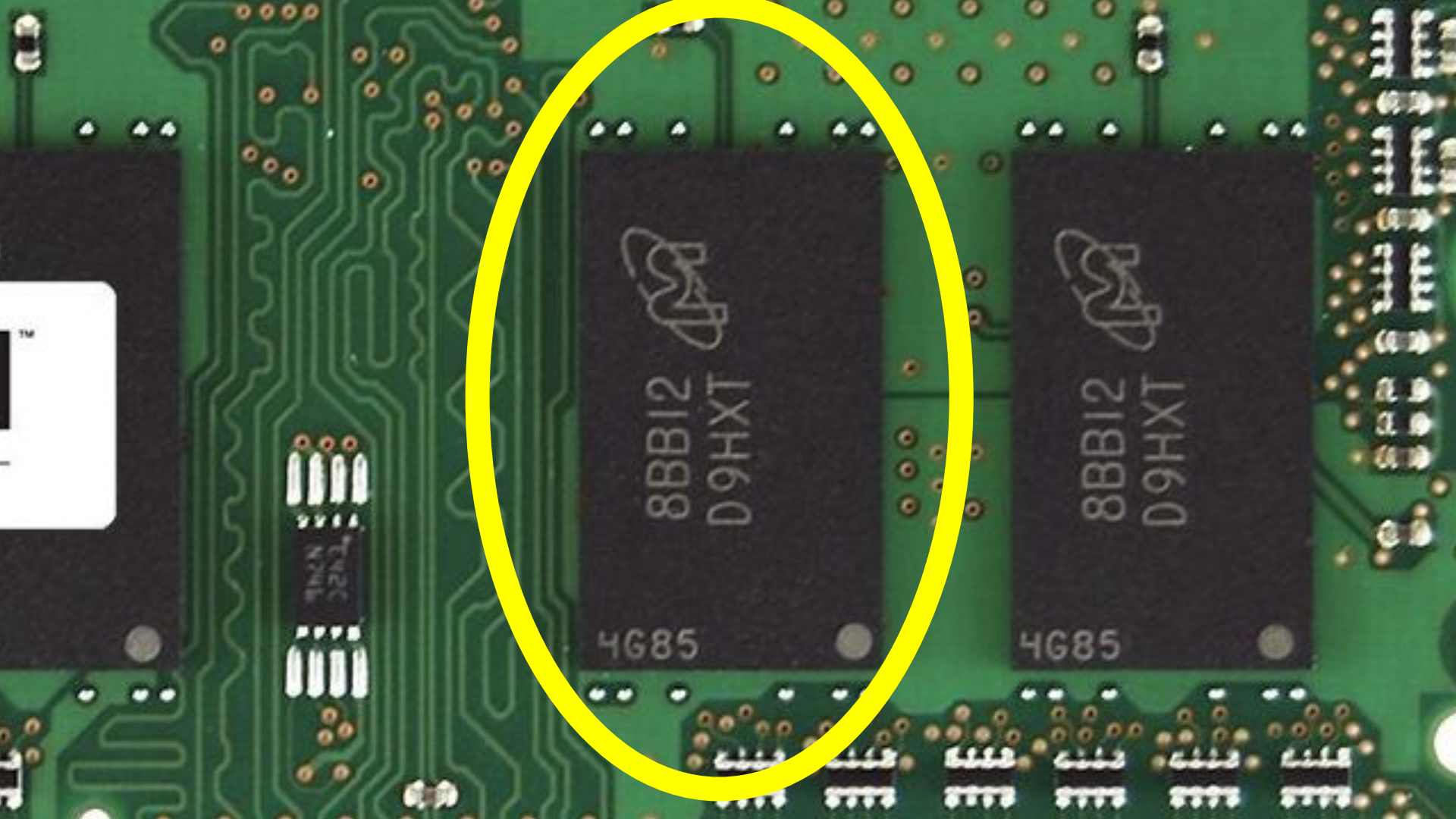
...

crucial
by Micron
47J6
4G85
8BB12
D9HXT
4G85
8BB12
D9HXT
4G85

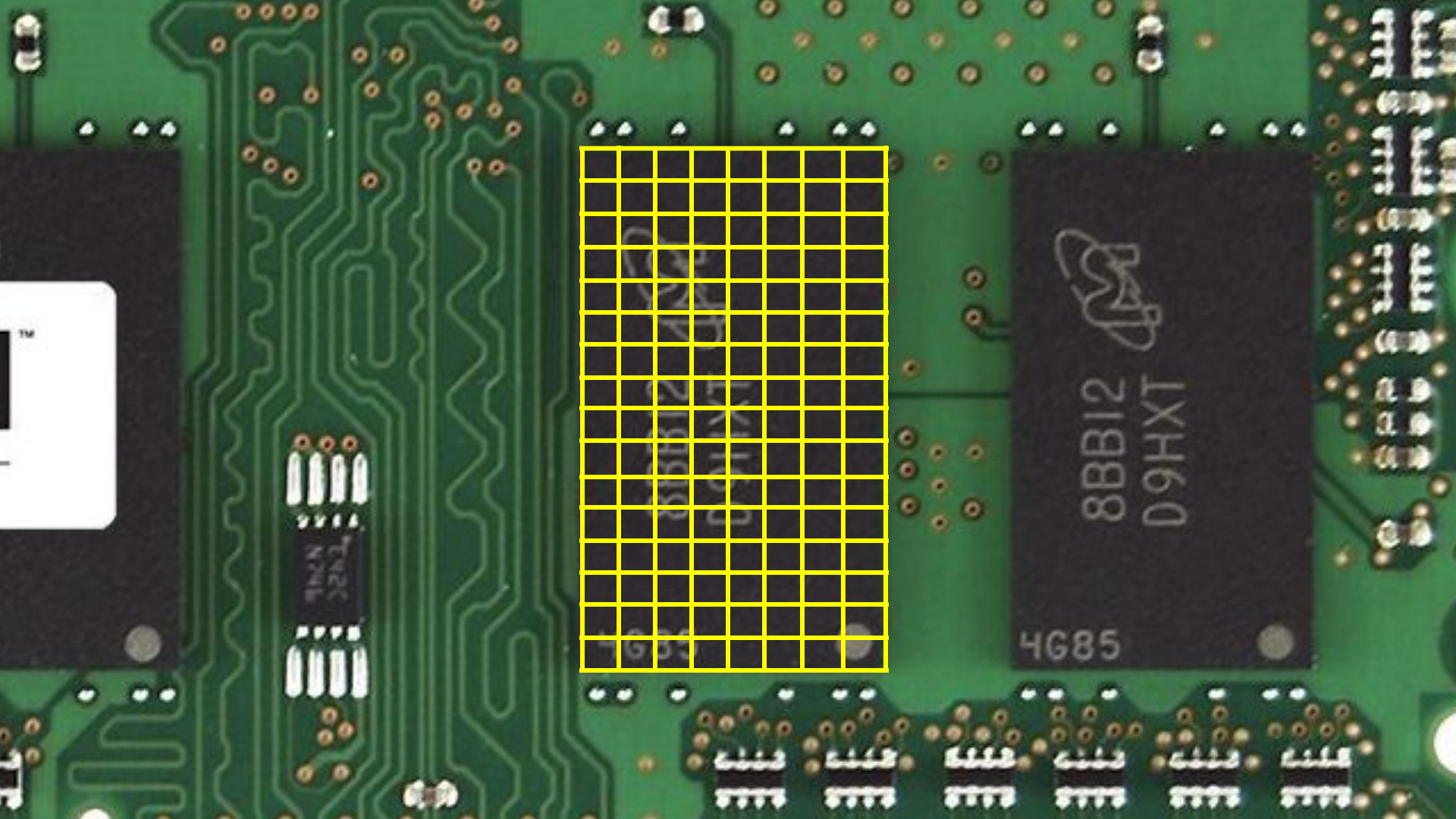
amazon.com

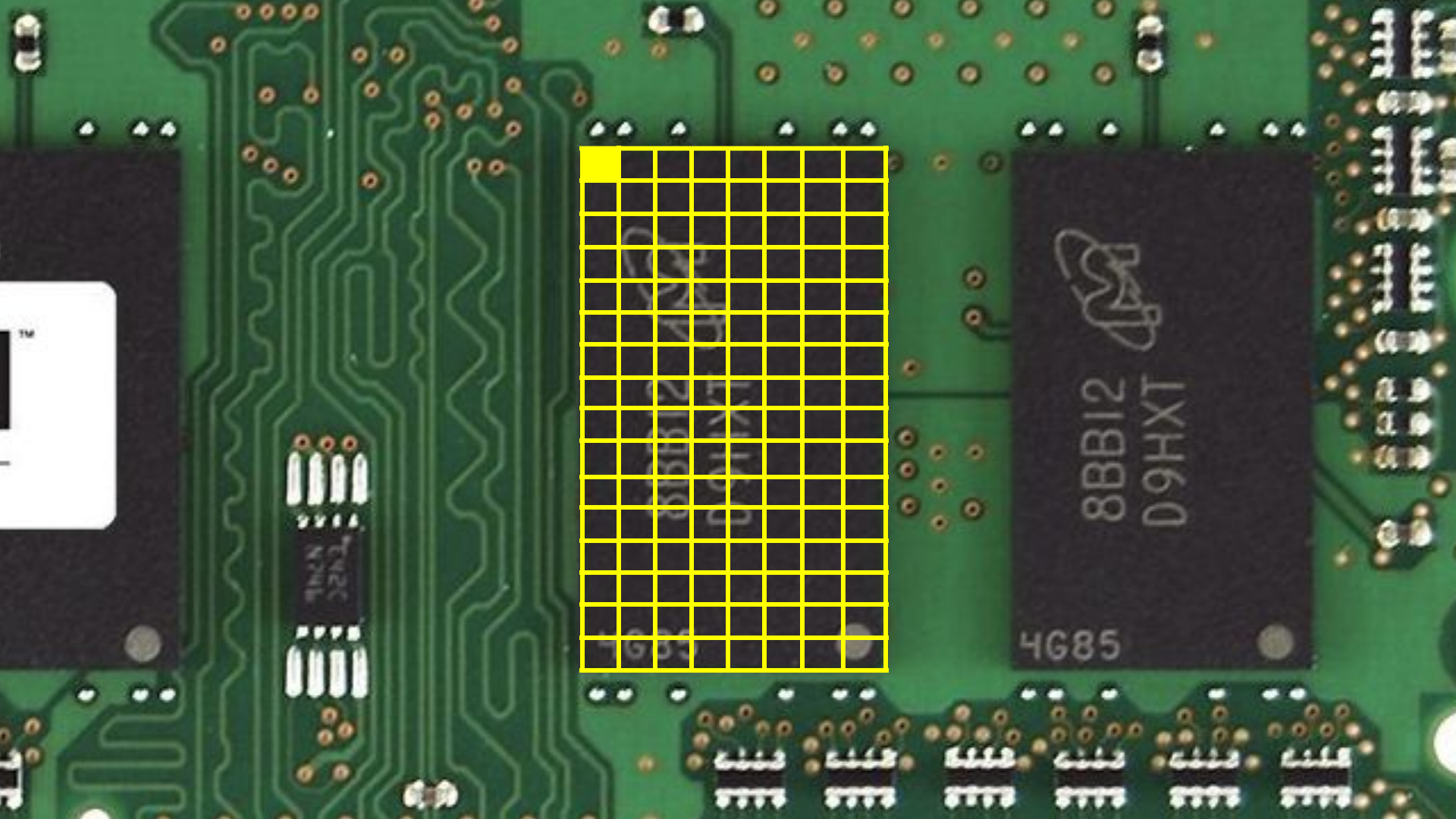
crucial
by Micron
47J6
4G85
8BB12
D9HXT
4G85
8BB12
D9HXT
4G85

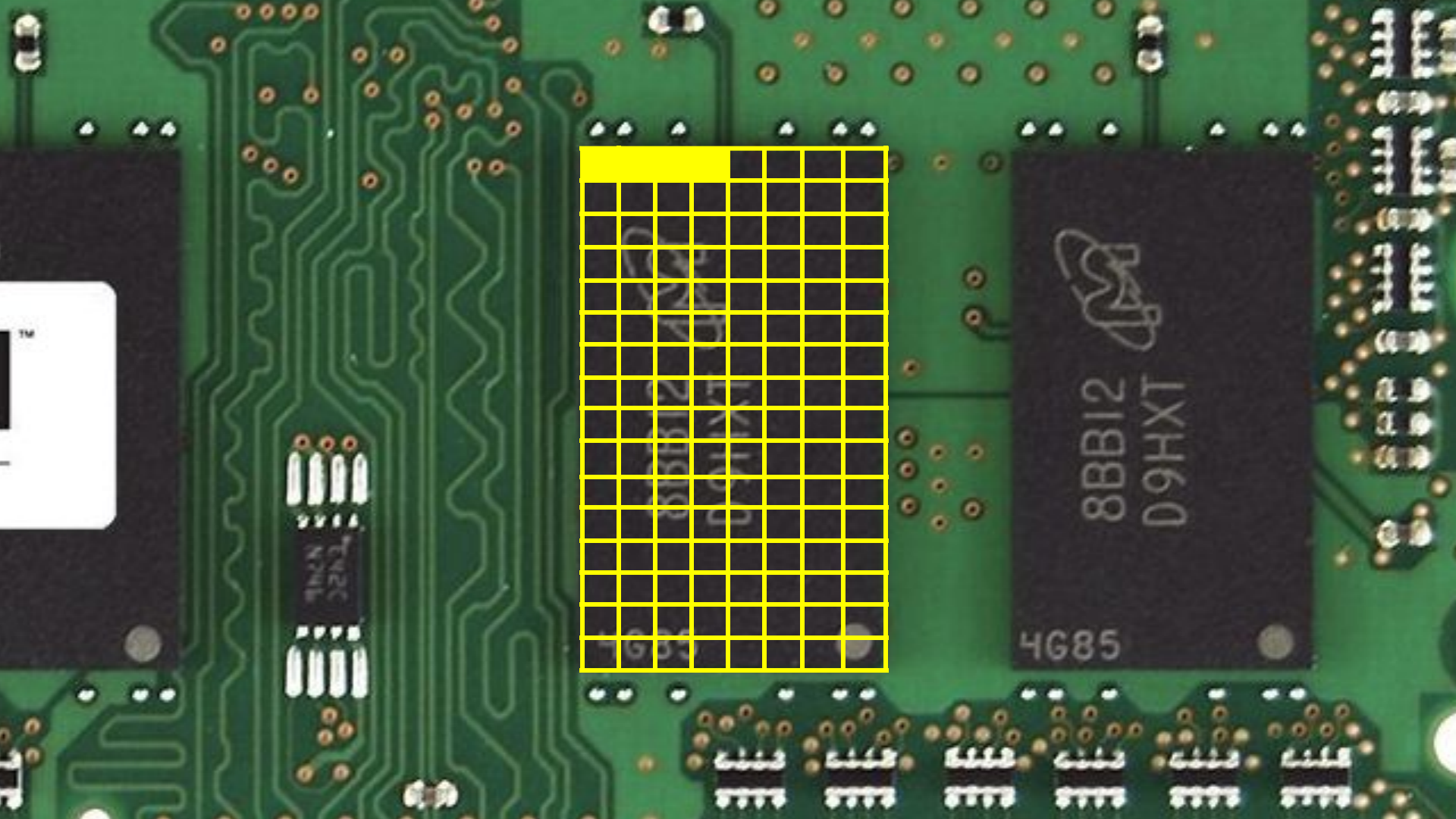
amazon.com

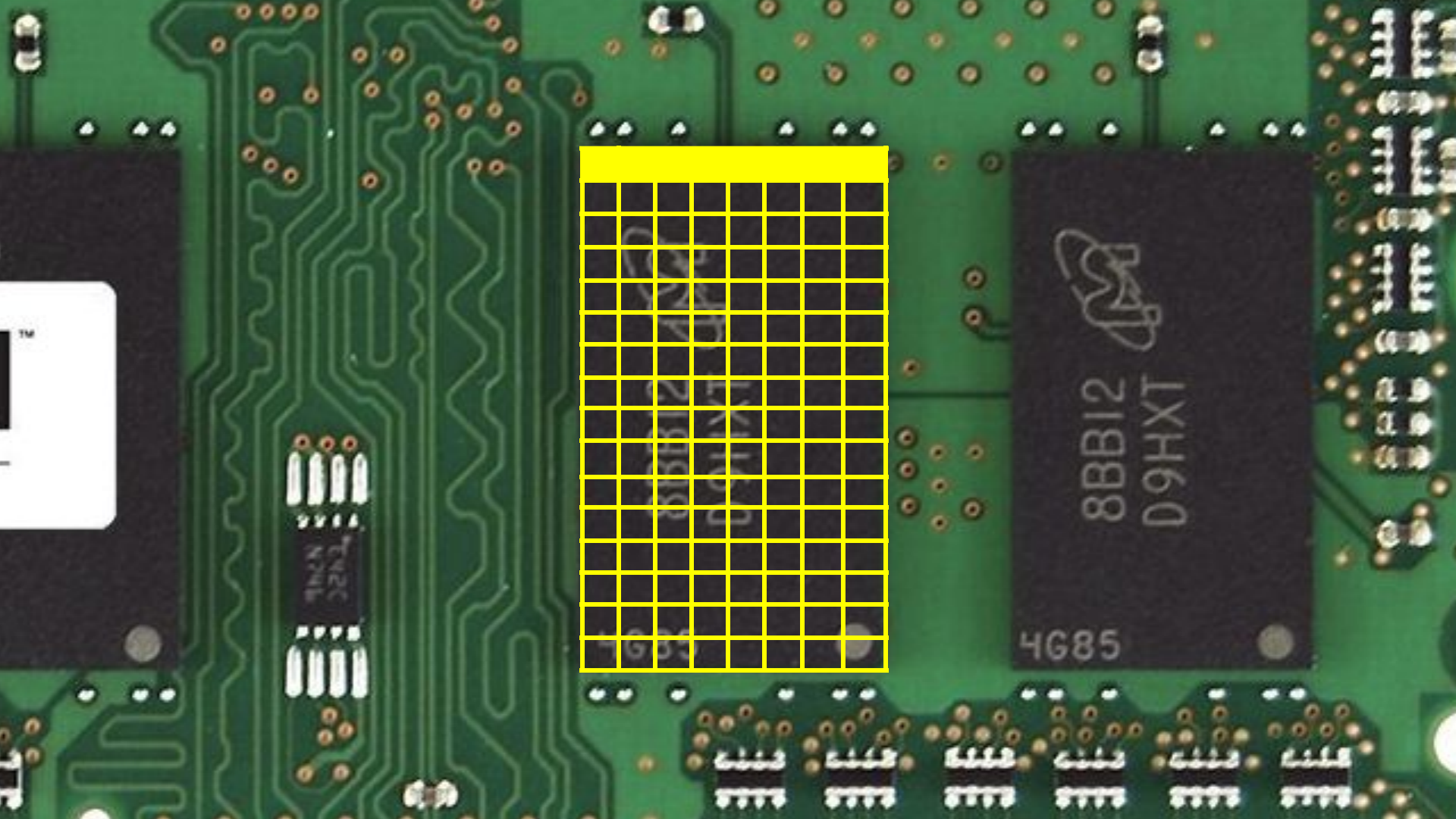

8BBI2
D9HXT
4G85
8BBI2
D9HXT
4G85

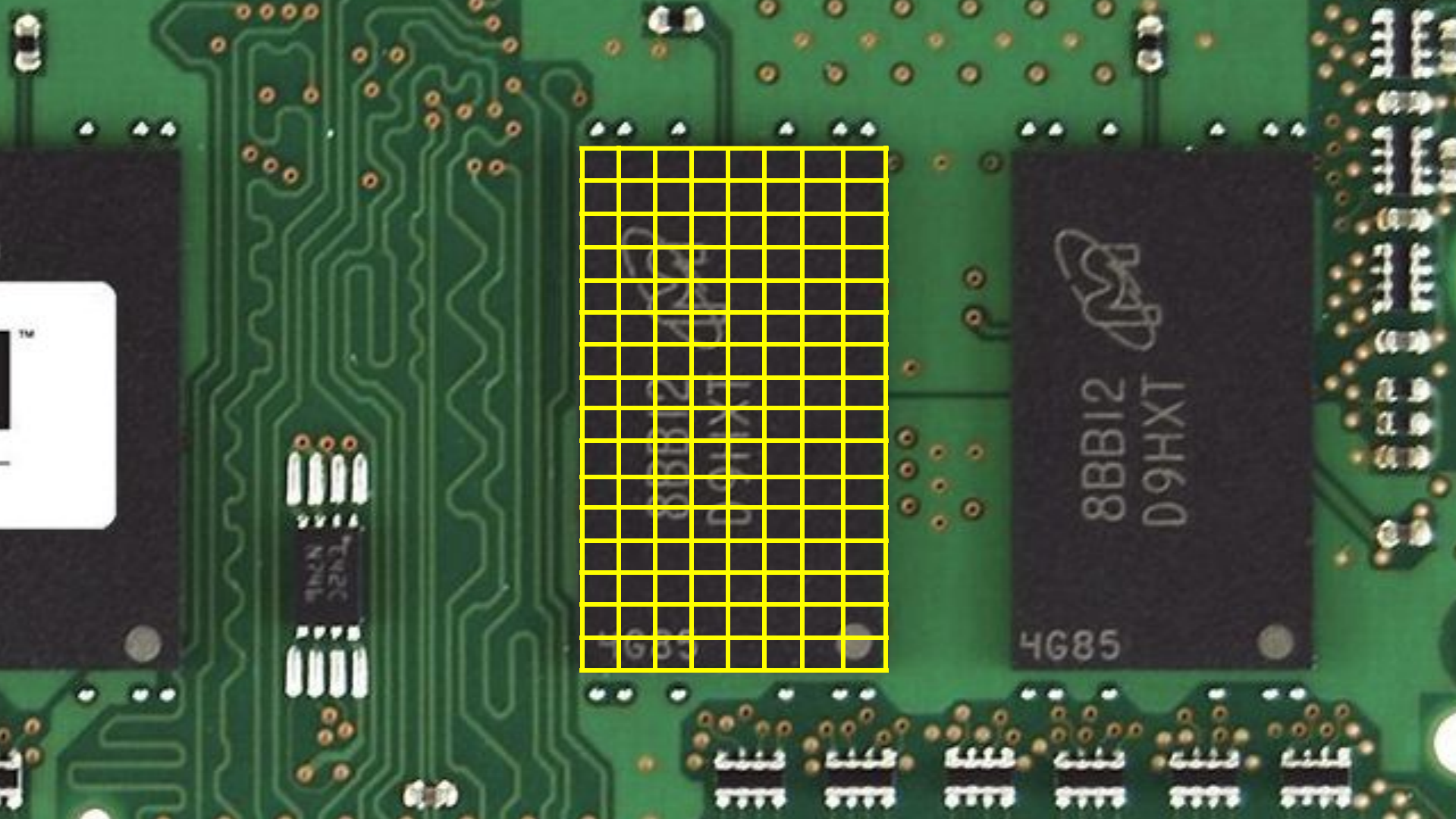
8BBI2
D9HXT
4G85
8BBI2
D9HXT
4G85

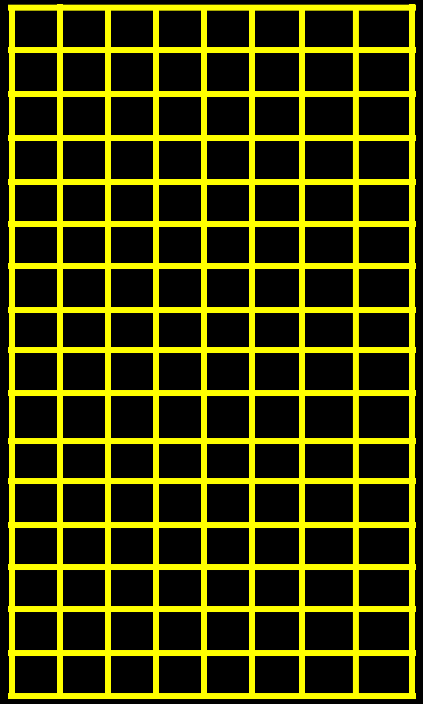

8BB12
D9HXT
4G85


8BB12
D9HXT
4G85


8BB12
D9HXT
4G85


8BB12
D9HXT
4G85


8BB12
D9HXT
4G85



```
int score1 = 72;

int score2 = 73;

int score3 = 33;
```

72

score1

72

score1

73

score2

72

score1

73

score2

33

score3

00000000000000000000000001001000

score1

00000000000000000000000001001001

score2

00000000000000000000000000100001

score3

```
int score1 = 72;

int score2 = 73;

int score3 = 33;
```

arrays

```
int scores[3];
```

```
int scores[3];

scores[0] = 72;

scores[1] = 73;

scores[2] = 33;
```

72

scores[0]

73

scores[1]

33

scores[2]

```
char c1 = 'H';

char c2 = 'I';

char c3 = '!';
```

H c1	I c2	! c3					

72 c1	73 c2	33 c3					

01001000 c1	01001001 c2	00100001 c3					

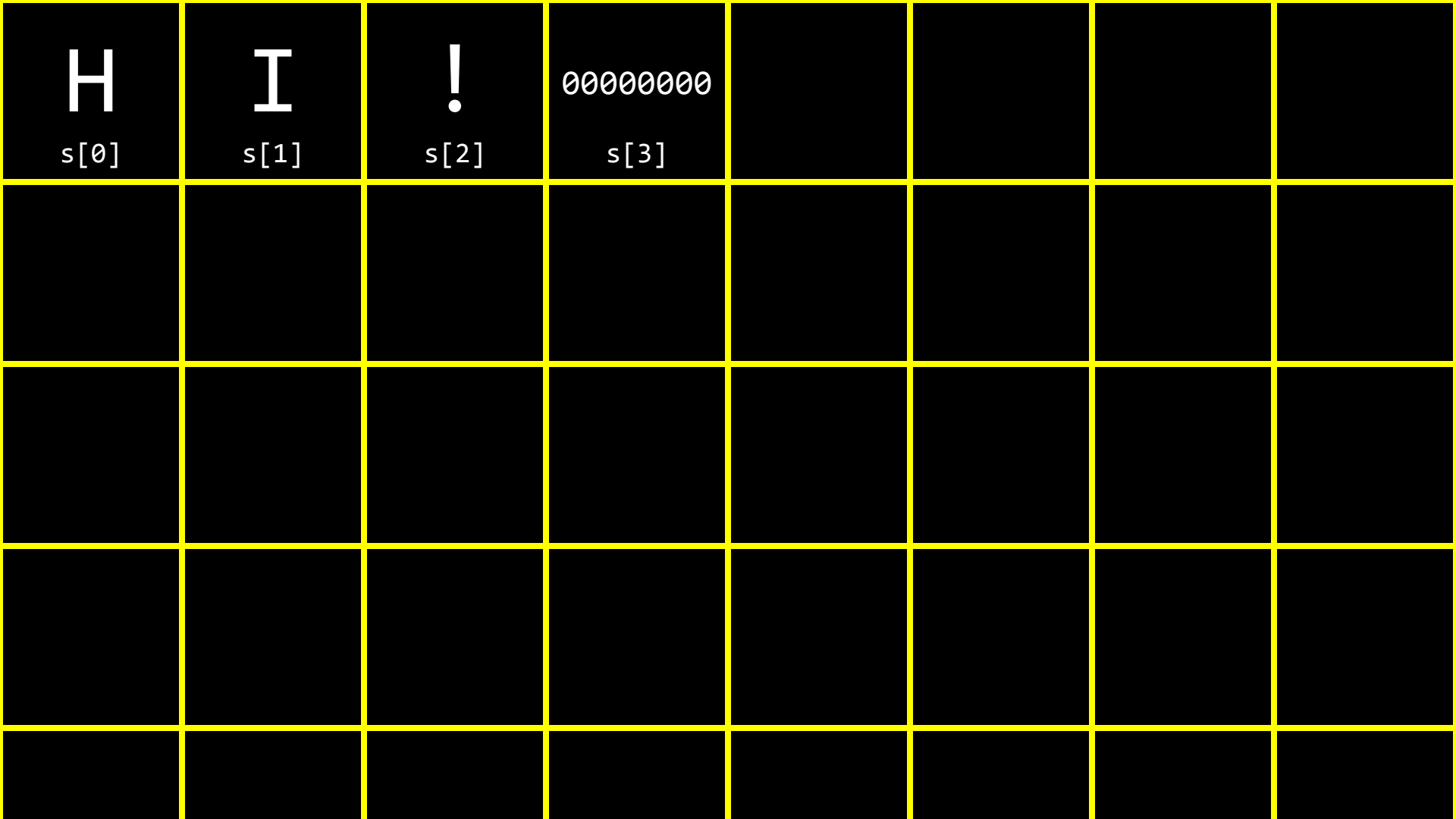
string

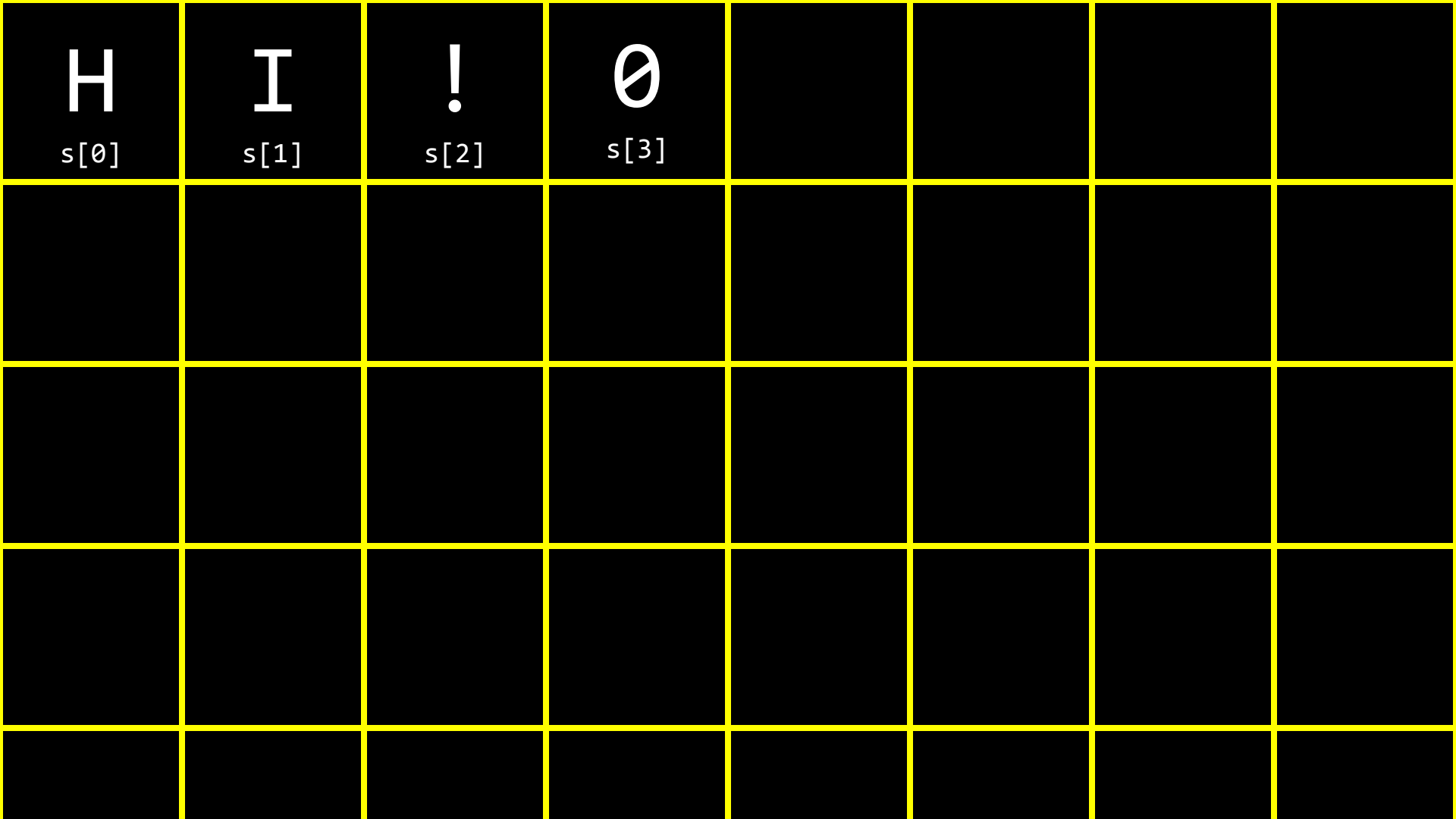
```
string s = "HI!";
```

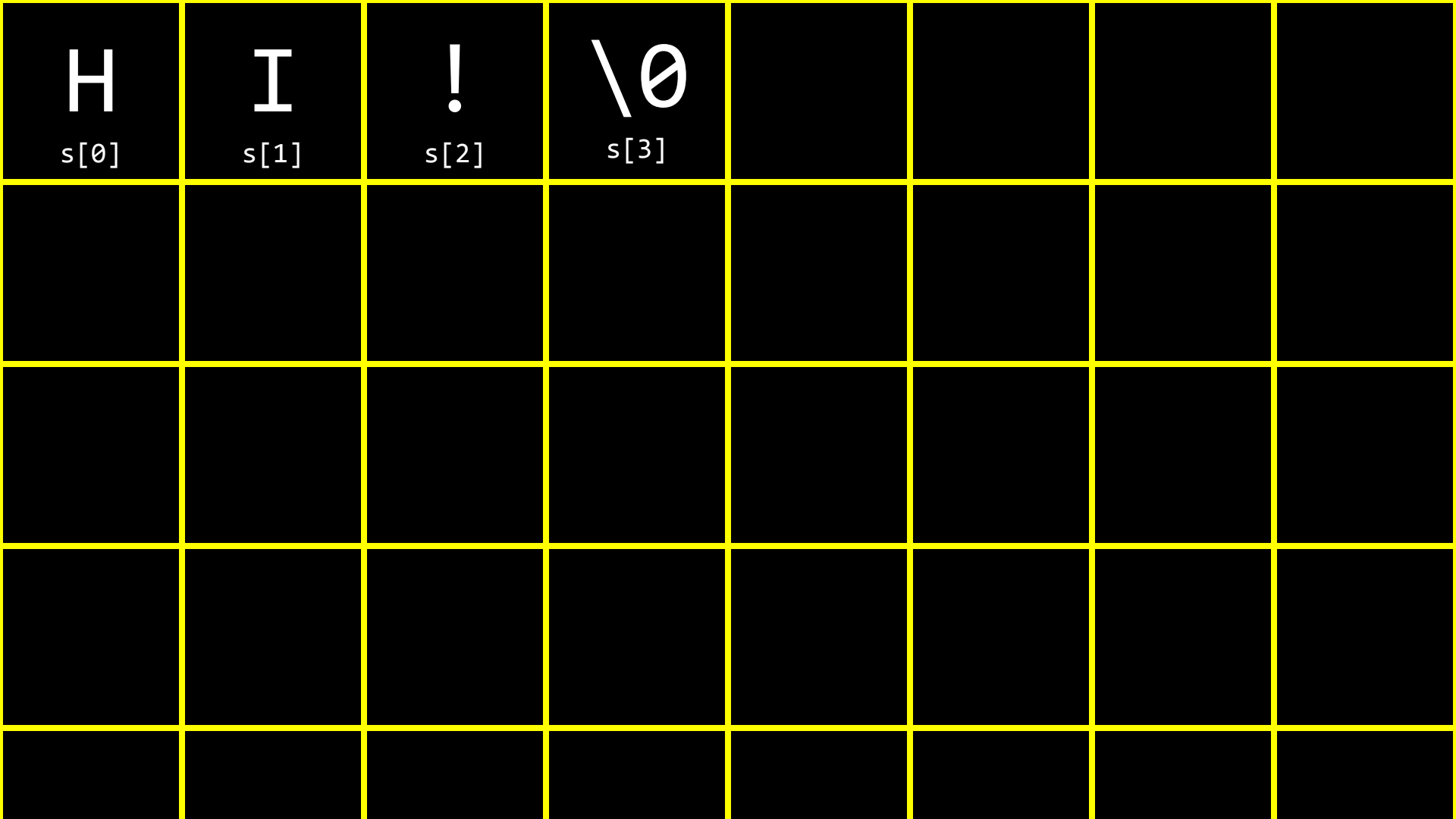
H I !

s

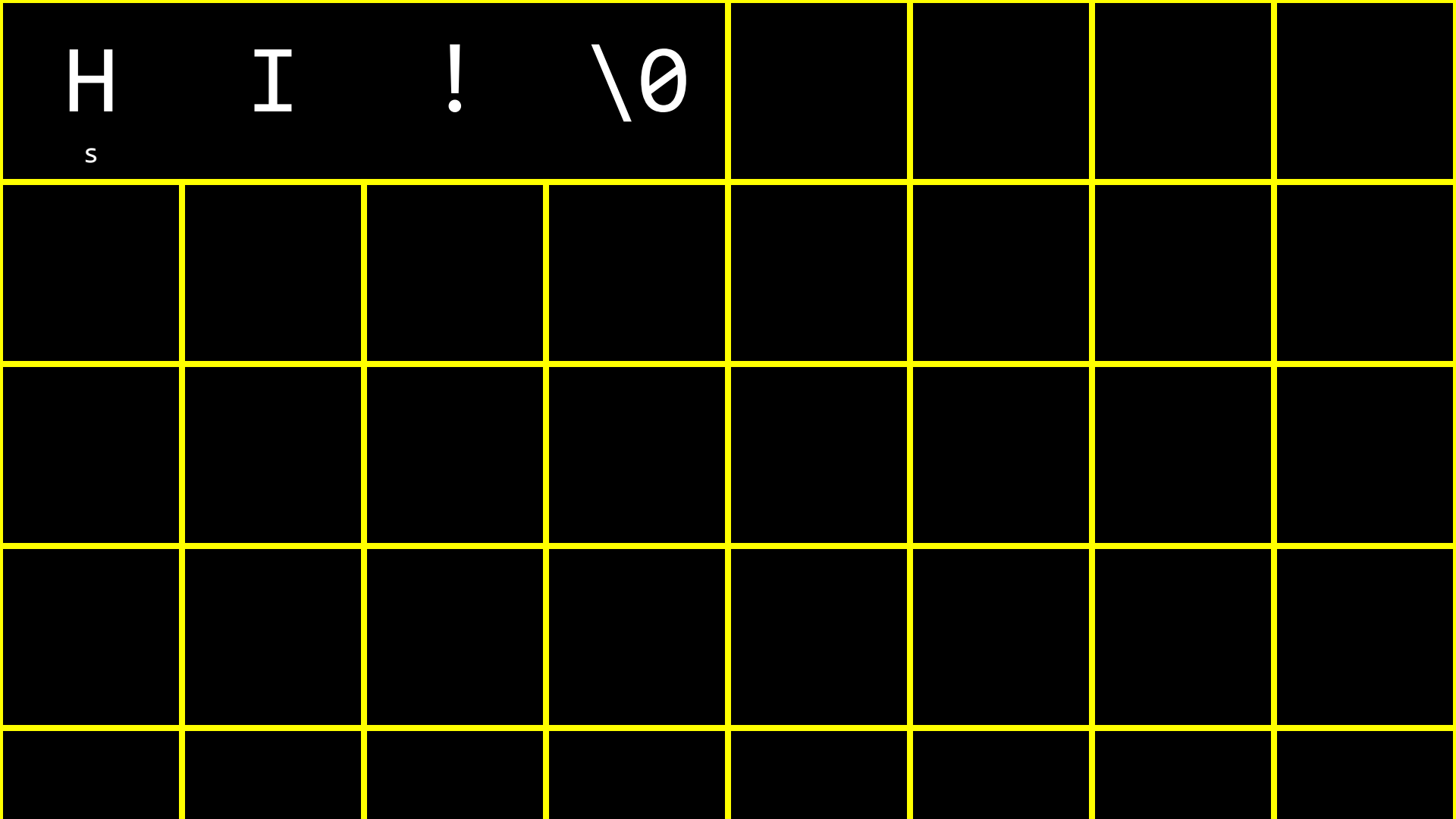
H s[0]	I s[1]	! s[2]					


H
s[0]
I
s[1]
!
s[2]
00000000
s[3]


H
s[0]
I
s[1]
!
s[2]
0
s[3]


H
s[0]
I
s[1]
!
s[2]
\0
s[3]

72 s[0]	73 s[1]	33 s[2]	0 s[3]				

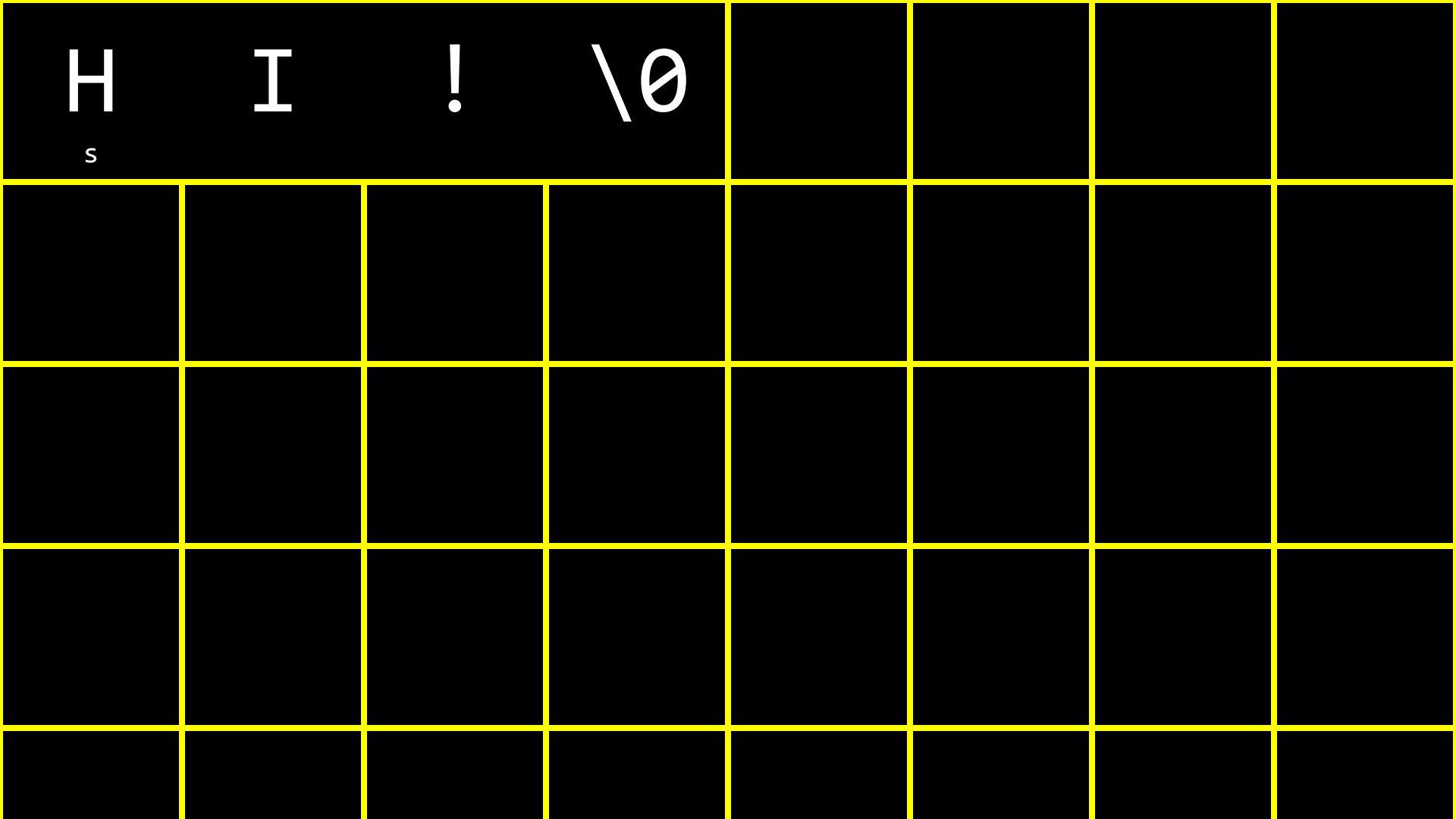

H
I
!
\0
s

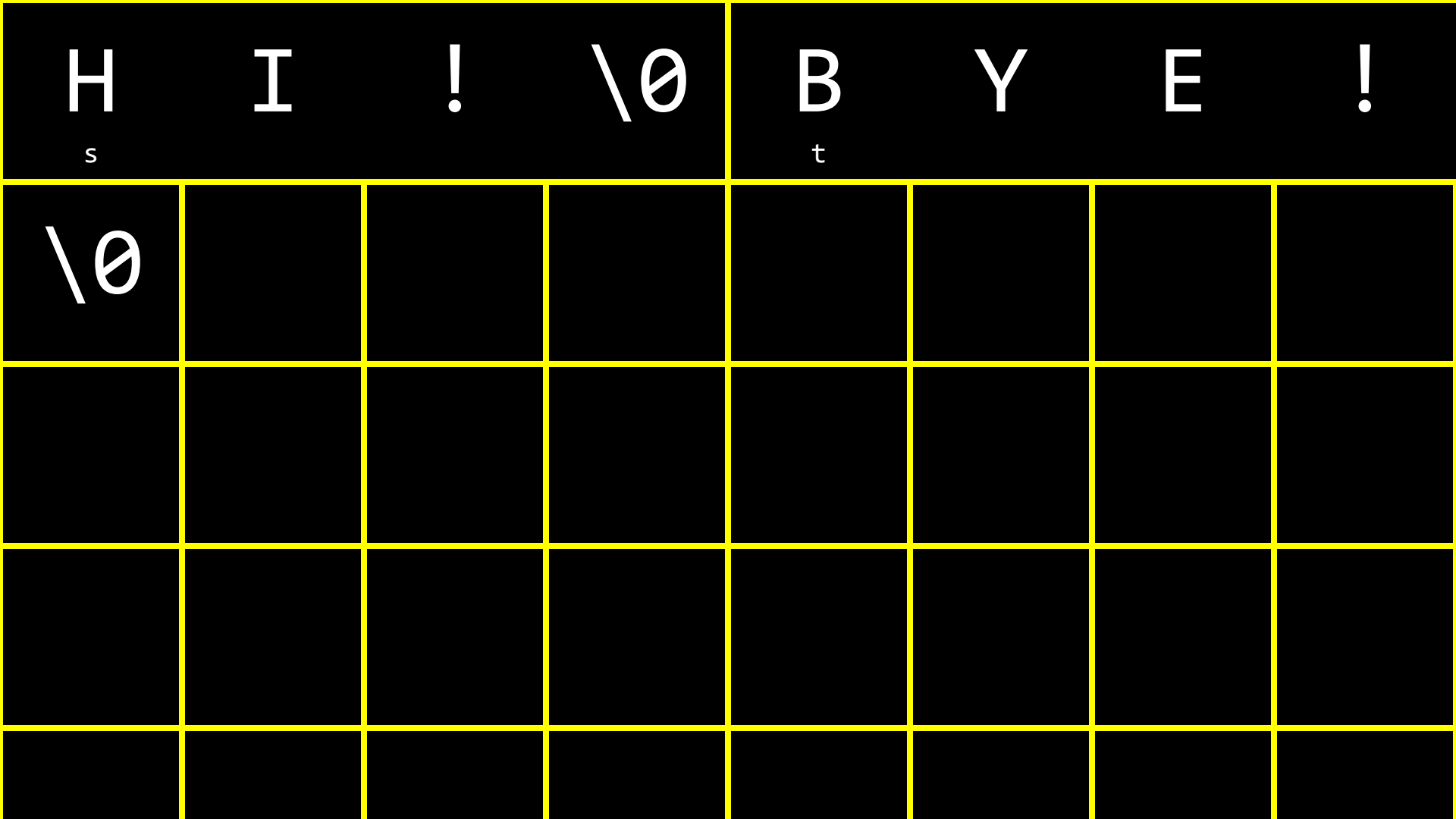
NUL

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

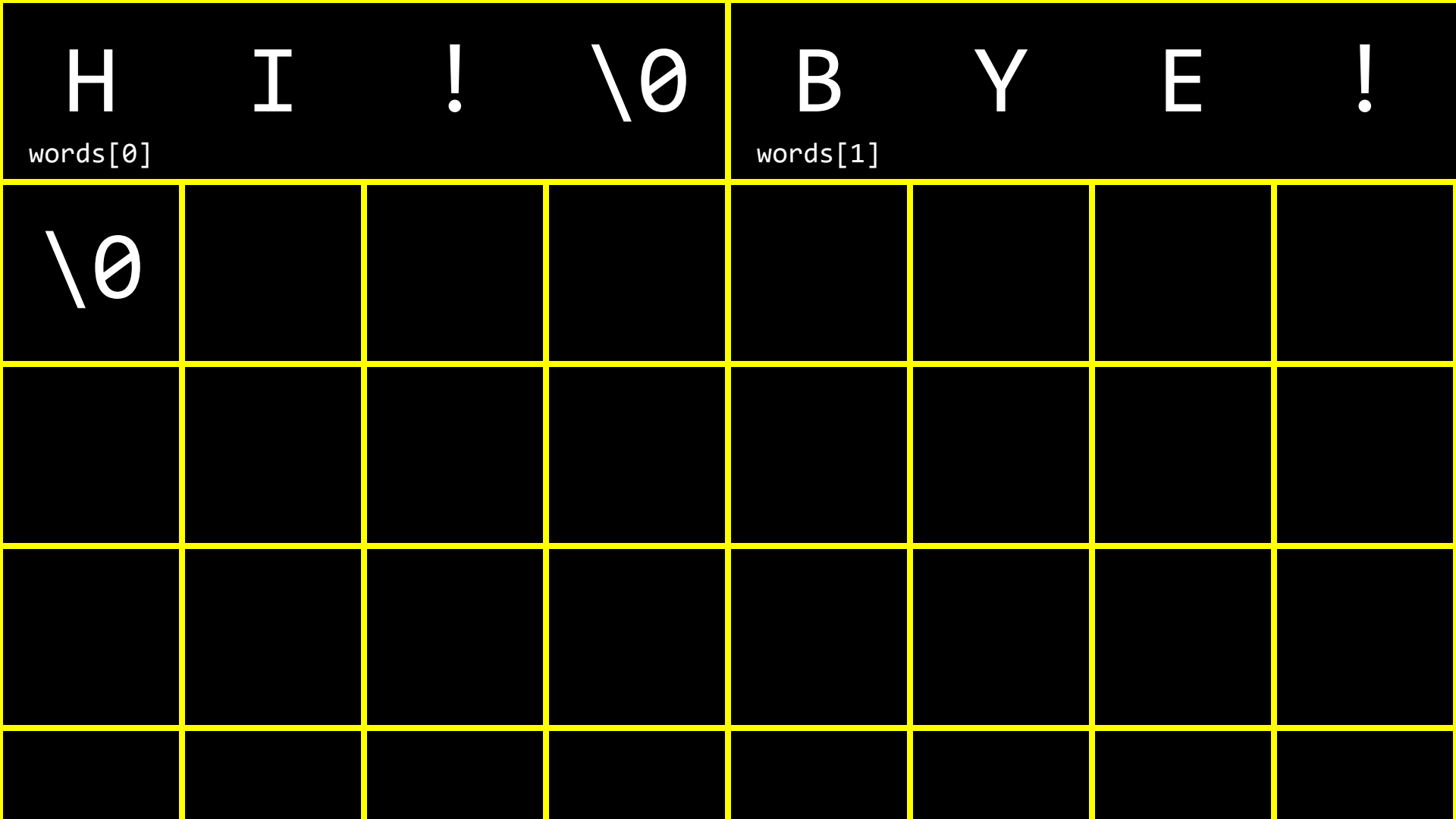
```
string s = "HI!";

string t = "BYE!";
```


H
I
!
\0
s


H I ! \0
s
B Y E !
t
\0

H	I	!	\0	B	Y	E	!
s[0]	s[1]	s[2]	s[3]	t[0]	t[1]	t[2]	t[3]
\0							
t[4]							

```
string words[2];

words[0] = "HI!";

words[1] = "BYE!";
```


H I ! \0
words[0]
B Y E !
words[1]
\0

H words[0][0]	I words[0][1]	! words[0][2]	\0 words[0][3]	B words[1][0]	Y words[1][1]	E words[1][2]	! words[1][3]
\0 words[1][4]							

string

string.h

manual.cs50.io/#string.h

strlen

ctype.h

manual.cs50.io/#ctype.h

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

main

```c
#include <stdio.h>

int main(void)
{
    ...
}
```

```
#include <stdio.h>

int main(void)
{
    ...
}
```

```
#include <stdio.h>

int main(int argc, string argv[])
{
    ...
}
```

ASCII art

cowsay

exit status

Zoom

An unknown error occurred

Error code: 1132

Report Problem

OK

404
This is not the web page you are looking for.

```
#include <stdio.h>

int main(int argc, string argv[])
{
    ...
}
```

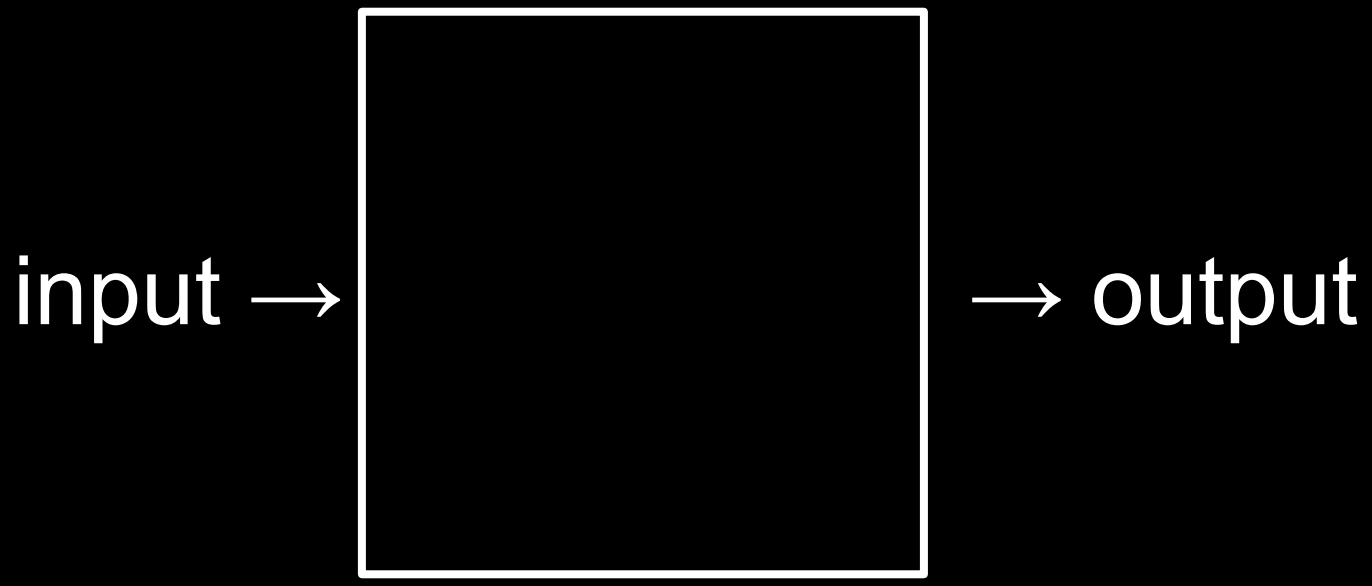
```
#include <stdio.h>

int main(int argc, string argv[])
{
    ...
}
```

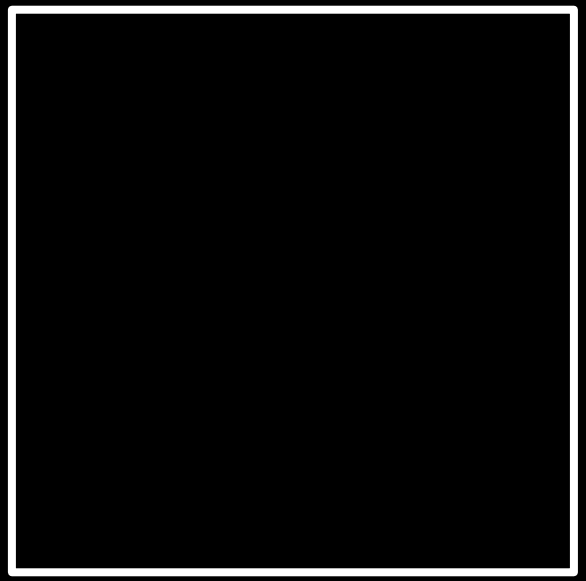
```c
#include <stdio.h>

int main(void)
{
    ...
}
```

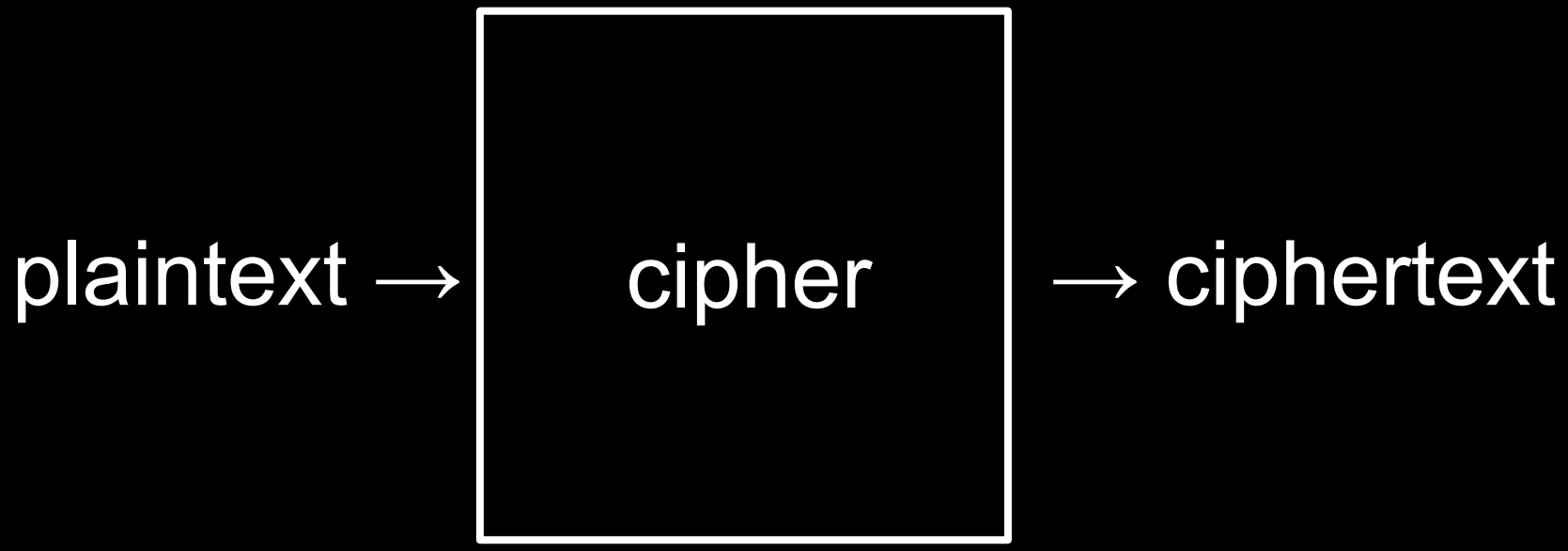
```
echo $?
```

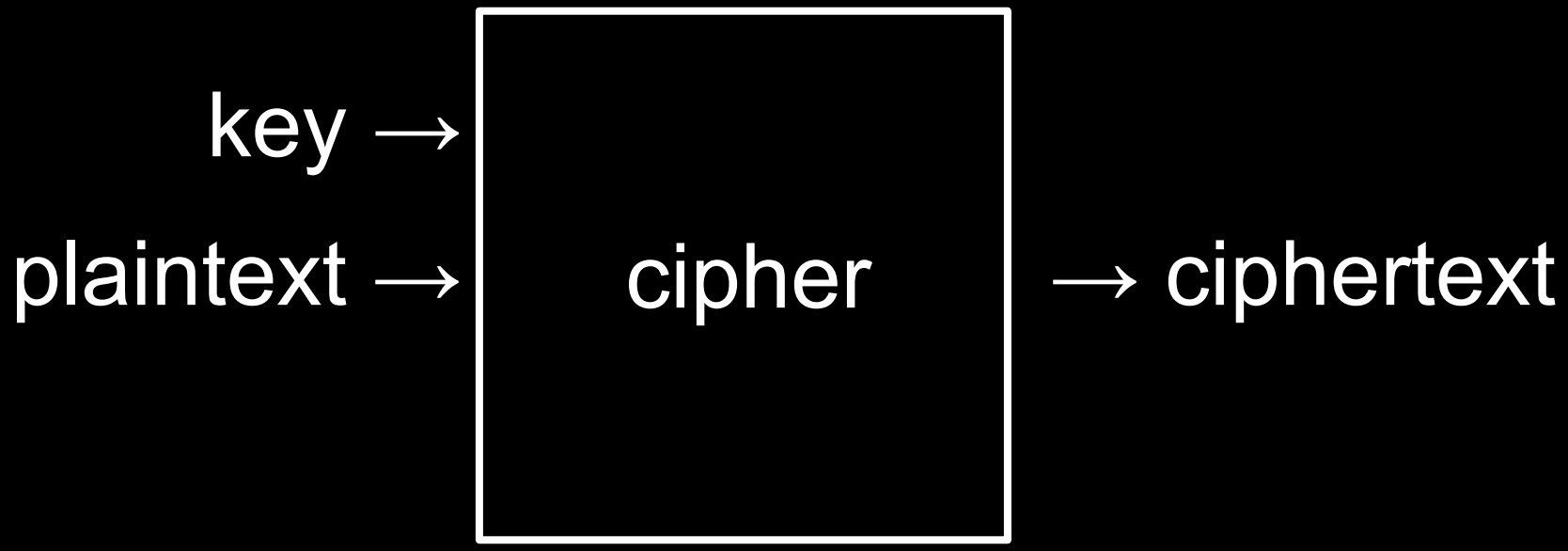
cryptography

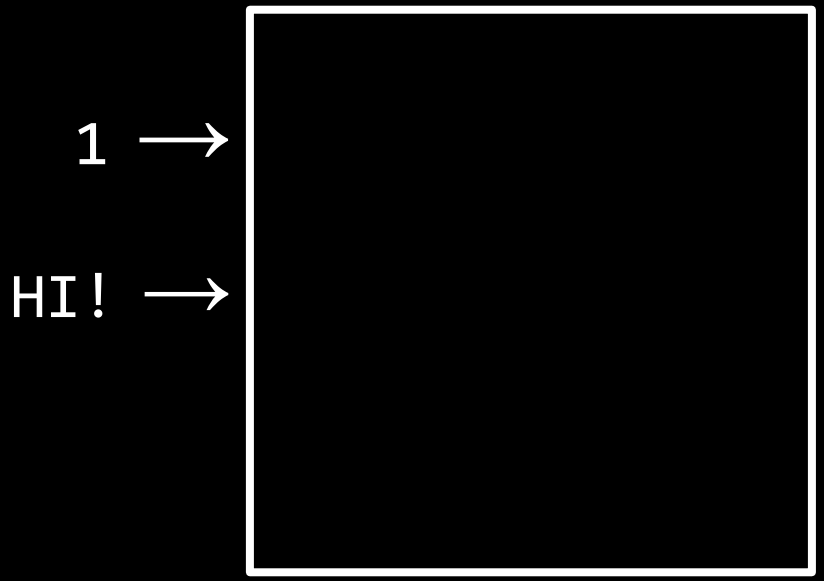
encryption

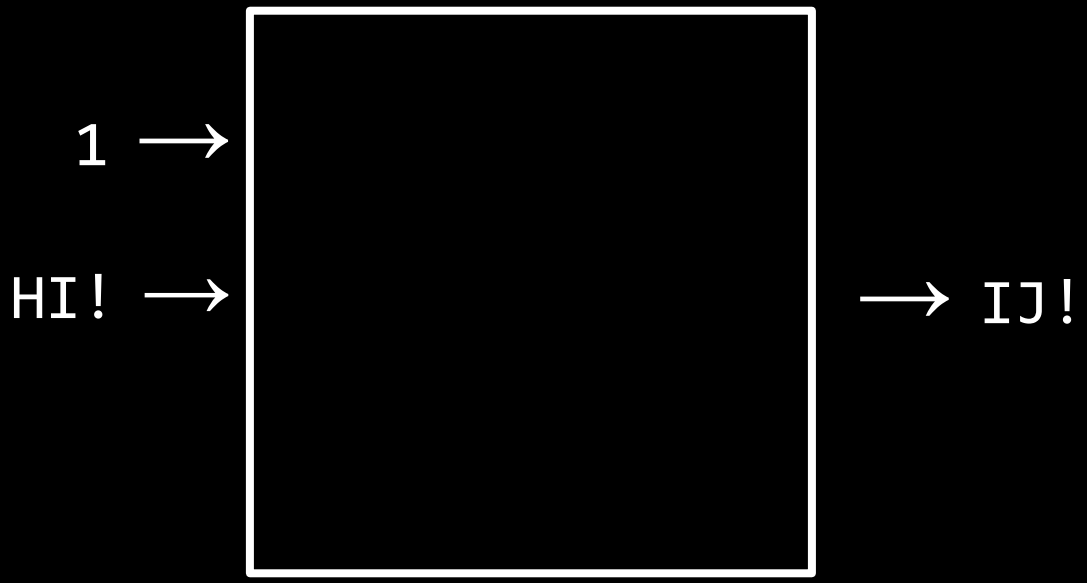

input →
→ output

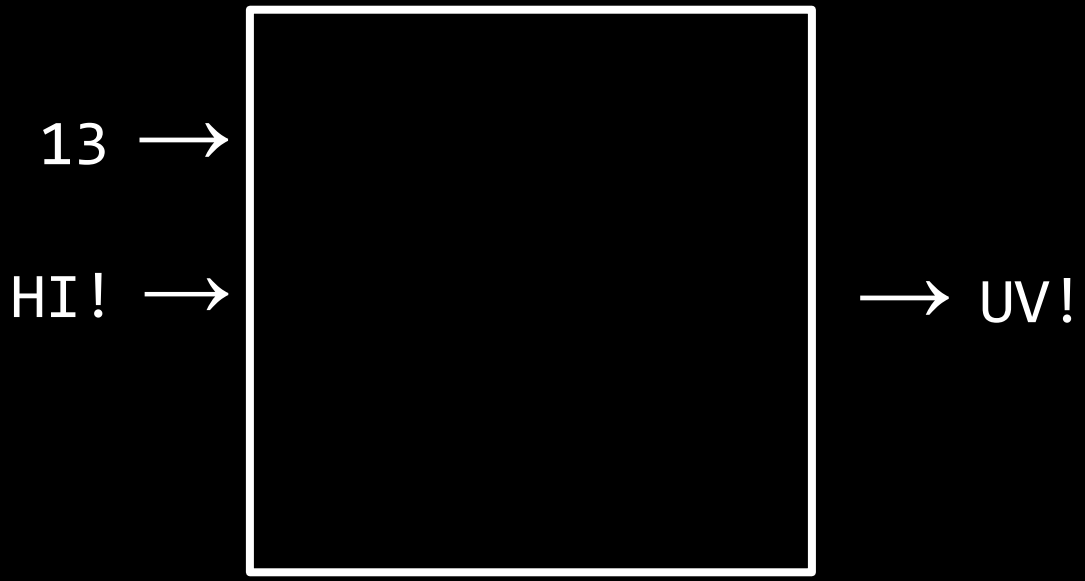
plaintext →  → ciphertext


plaintext →
cipher
→ ciphertext


key →
plaintext →
cipher
→ ciphertext

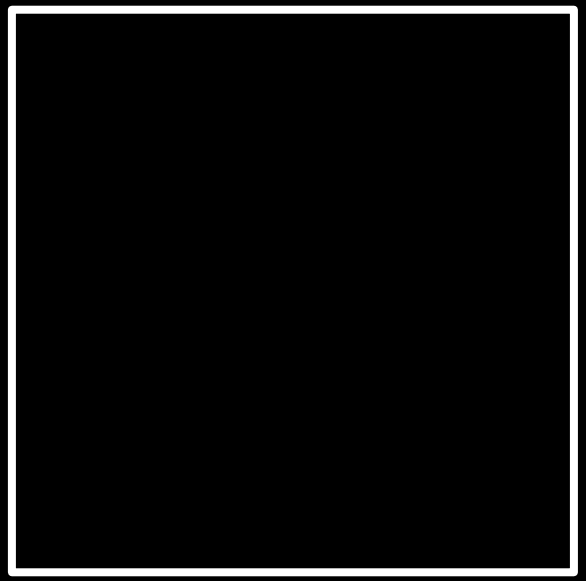

1
HI!


1
HI!
IJ!


13
HI!
UV!

13 →

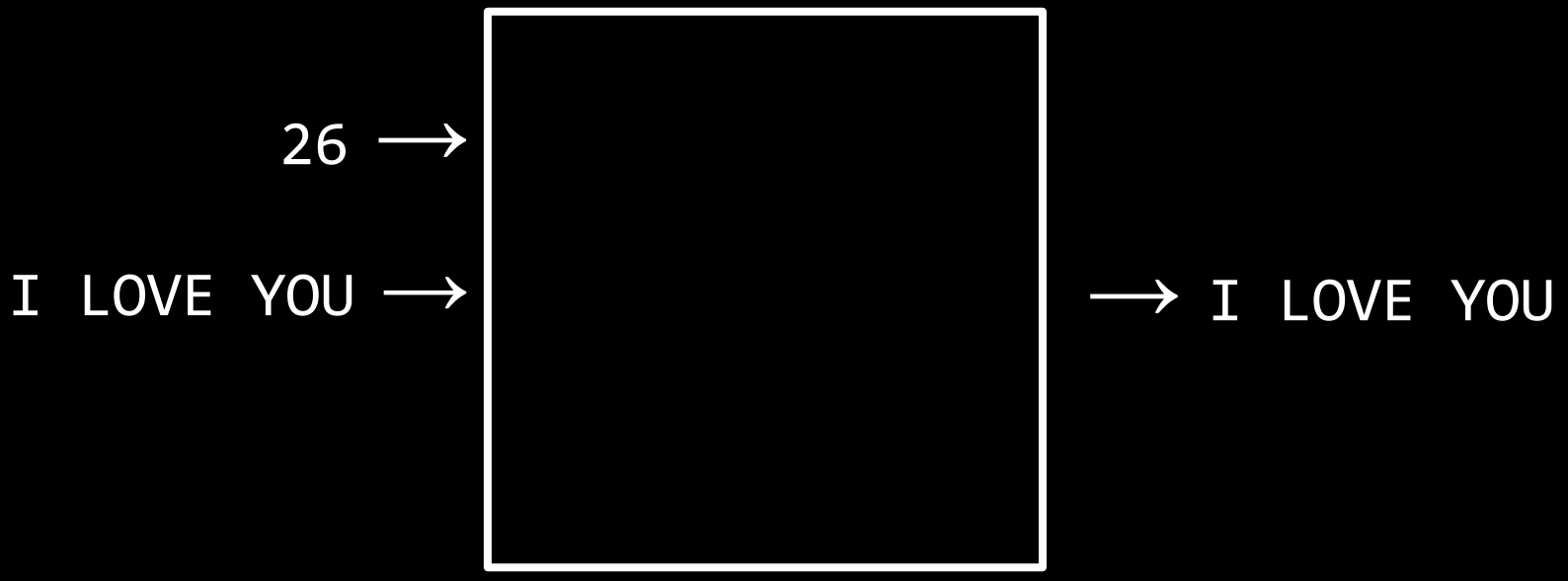
I LOVE YOU →



→ V YBIR LBH

26 →

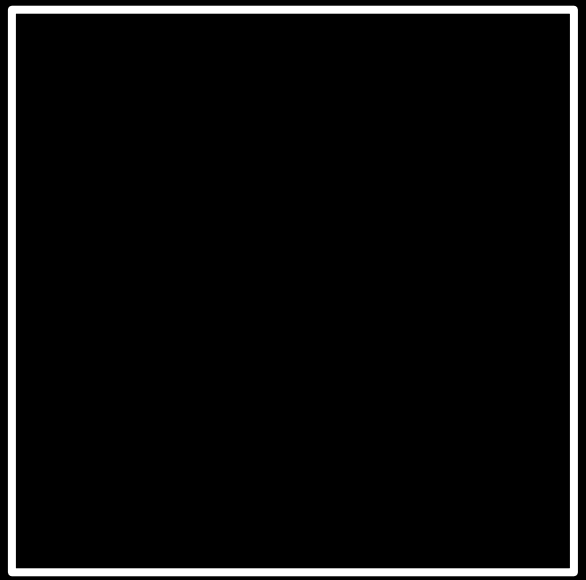
I LOVE YOU →


26
I LOVE YOU
I LOVE YOU

decryption

-1 →

UIJT XBT DT50 →



U I J T X B T D T 5 0

T I J T X B T D T 5 0

T H J T X B T D T 5 0

T H I T X B T D T 5 0

T H I S X B T D T 5 0

T H I S W B T D T 5 0

T H I S W A T D T 5 0

T H I S W A S D T 5 0

T H I S W A S C T 5 0

T H I S W A S C S 5 0

CS50
ddb

This is CS50